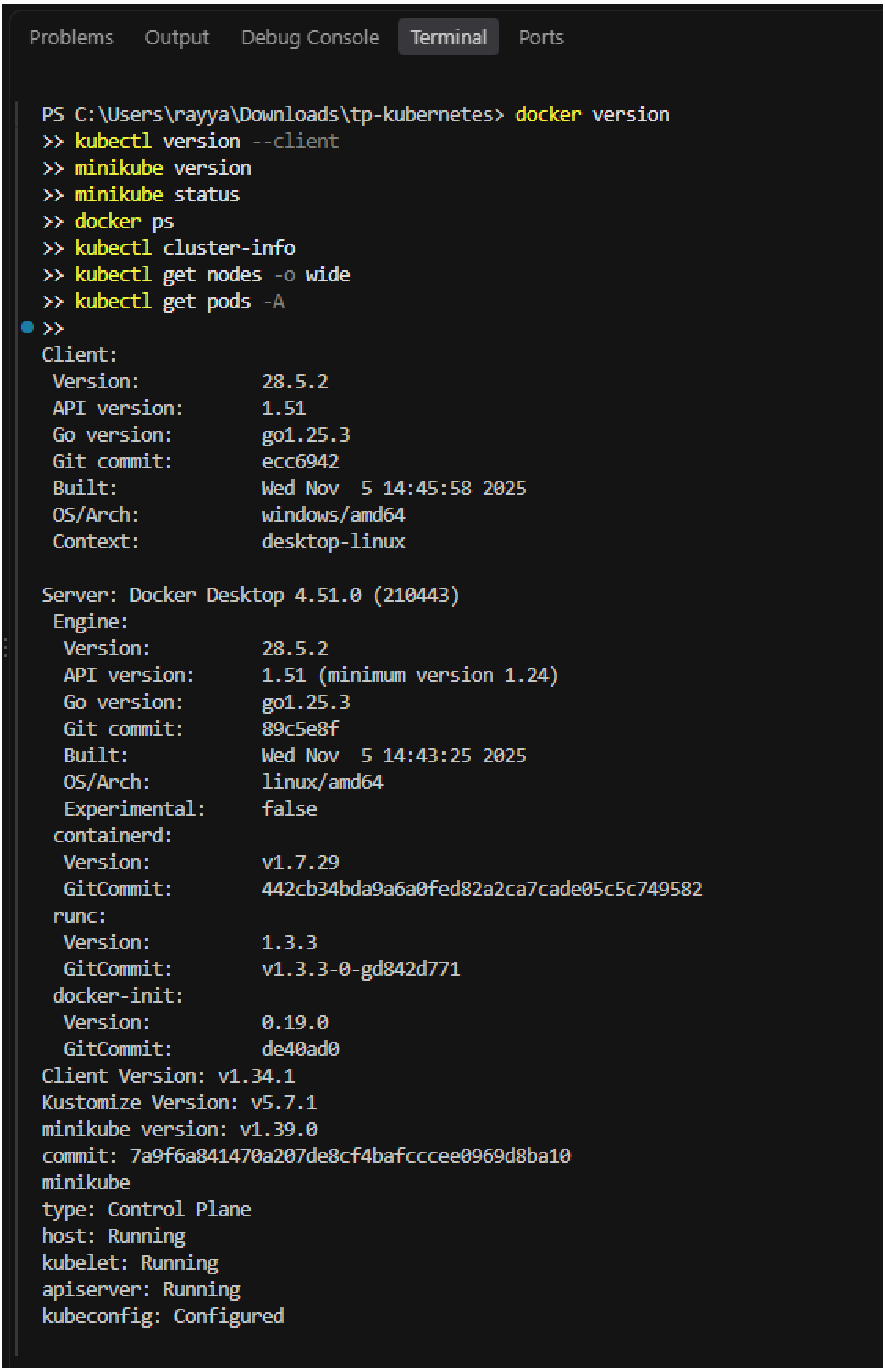
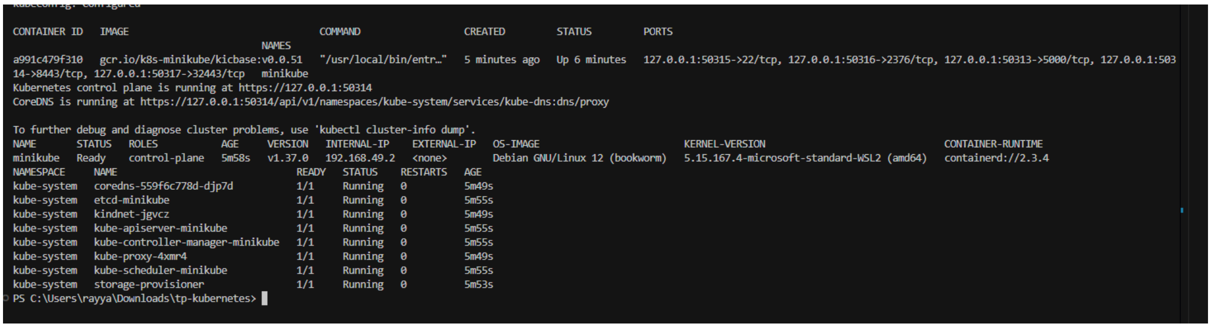
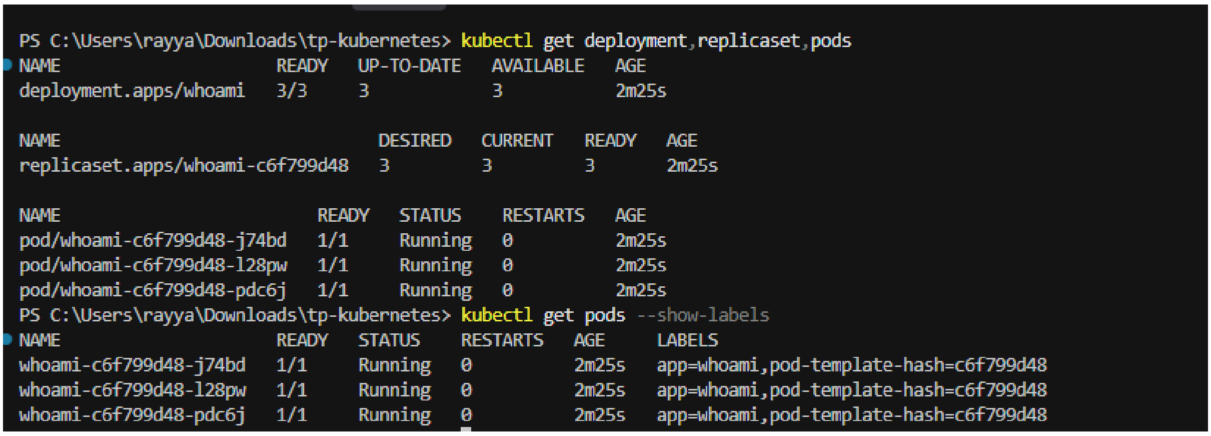


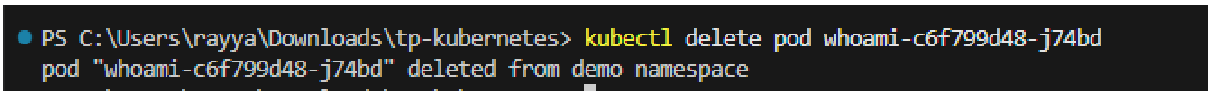
```
PS C:\Users\rayya\Downloads\tp-kubernetes> docker version
>> kubectl version --client
>> minikube version
>> minikube status
>> docker ps
>> kubectl cluster-info
>> kubectl get nodes -o wide
>> kubectl get pods -A
>>
Client:
 Version:           28.5.2
 API version:       1.51
 Go version:        go1.25.3
 Git commit:        ecc6942
 Built:             Wed Nov  5 14:45:58 2025
 OS/Arch:           windows/amd64
 Context:           desktop-linux

Server: Docker Desktop 4.51.0 (210443)
 Engine:
  Version:          28.5.2
  API version:      1.51 (minimum version 1.24)
  Go version:       go1.25.3
  Git commit:       89c5e8f
  Built:            Wed Nov  5 14:43:25 2025
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          v1.7.29
  GitCommit:        442cb34bda9a6a0fed82a2ca7cade05c5c749582
 runc:
  Version:          1.3.3
  GitCommit:        v1.3.3-0-gd842d771
 docker-init:
  Version:          0.19.0
  GitCommit:        de40ad0
Client Version: v1.34.1
Kustomize Version: v5.7.1
minikube version: v1.39.0
commit: 7a9f6a841470a207de8cf4bafcccee0969d8ba10
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```



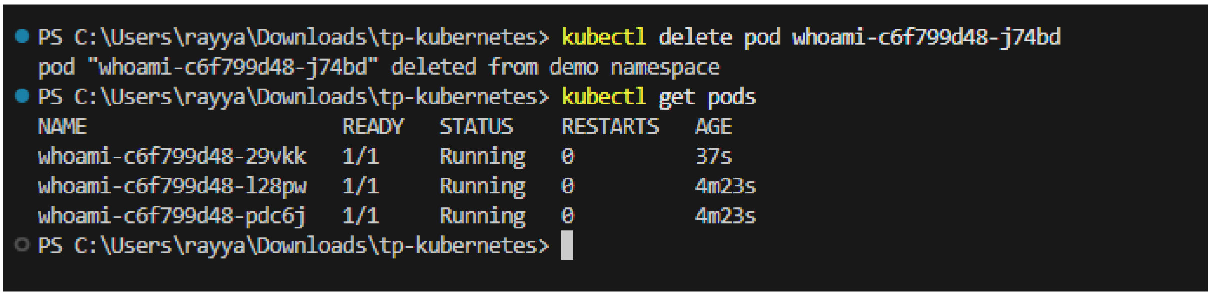
```
CONTAINER ID   IMAGE                                 COMMAND                  CREATED         STATUS          PORTS
                                      NAMES
a991c479f310   gcr.io/k8s-minikube/kicbase:v0.0.51   "/usr/local/bin/entr…"   5 minutes ago   Up 6 minutes   127.0.0.1:50315->22/tcp, 127.0.0.1:50316->2376/tcp, 127.0.0.1:50313->5000/tcp, 127.0.0.1:503
14->8443/tcp, 127.0.0.1:50317->32443/tcp   minikube
Kubernetes control plane is running at https://127.0.0.1:50314
CoreDNS is running at https://127.0.0.1:50314/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
NAME       STATUS   ROLES           AGE     VERSION   INTERNAL-IP    EXTERNAL-IP   OS-IMAGE                         KERNEL-VERSION                          CONTAINER-RUNTIME
minikube   Ready    control-plane   5m58s   v1.37.0   192.168.49.2   <none>        Debian GNU/Linux 12 (bookworm)   5.15.167.4-microsoft-standard-WSL2 (amd64)   containerd://2.3.4
NAMESPACE     NAME                               READY   STATUS    RESTARTS   AGE
kube-system   coredns-559f6c778d-djp7d           1/1     Running   0          5m49s
kube-system   etcd-minikube                      1/1     Running   0          5m55s
kube-system   kindnet-jgvcz                      1/1     Running   0          5m49s
kube-system   kube-apiserver-minikube            1/1     Running   0          5m55s
kube-system   kube-controller-manager-minikube   1/1     Running   0          5m55s
kube-system   kube-proxy-4xmr4                   1/1     Running   0          5m49s
kube-system   kube-scheduler-minikube            1/1     Running   0          5m55s
kube-system   storage-provisioner                1/1     Running   0          5m53s
PS C:\Users\rayya\Downloads\tp-kubernetes> 
```



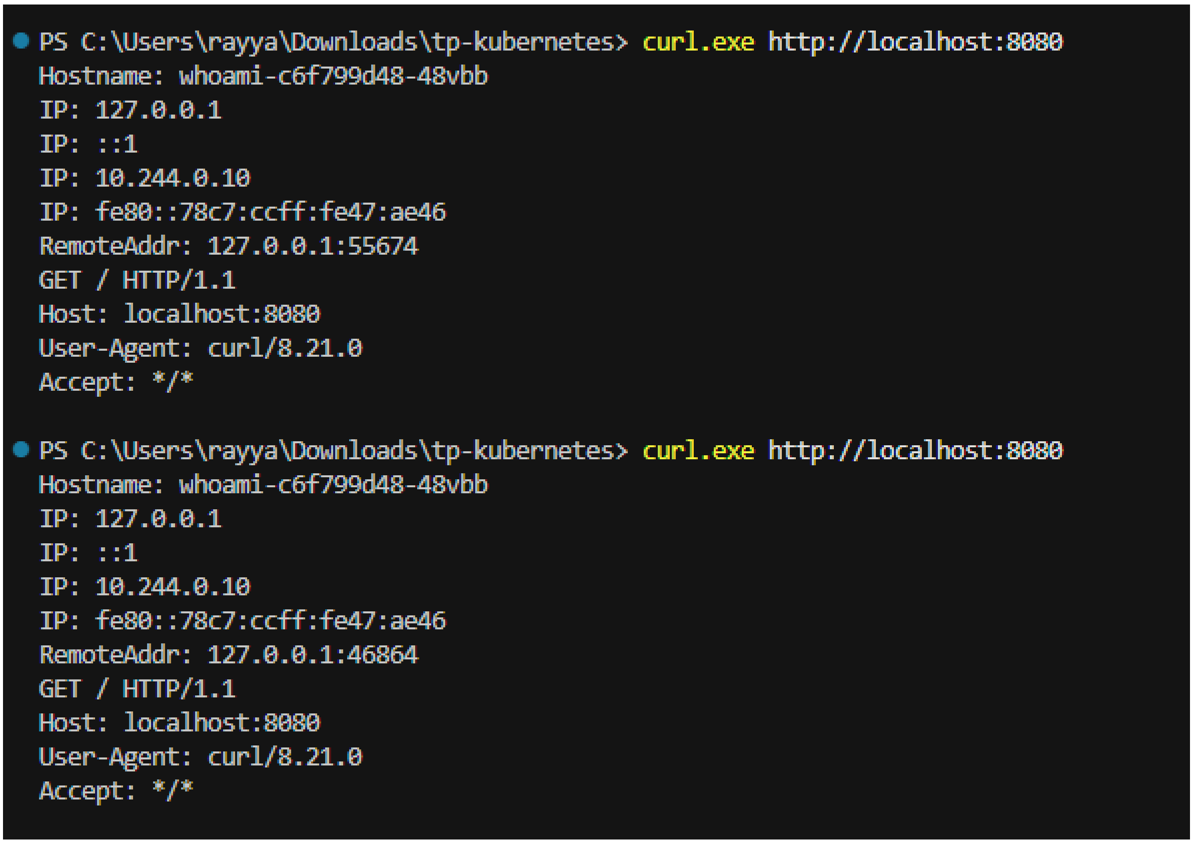
```
PS C:\Users\rayya\Downloads\tp-kubernetes> kubectl get deployment,replicaset,pods
NAME                     READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/whoami   3/3     3            3           2m25s

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/whoami-c6f799d48   3         3         3       2m25s

NAME                         READY   STATUS    RESTARTS   AGE
pod/whoami-c6f799d48-j74bd   1/1     Running   0          2m25s
pod/whoami-c6f799d48-l28pw   1/1     Running   0          2m25s
pod/whoami-c6f799d48-pdc6j   1/1     Running   0          2m25s
PS C:\Users\rayya\Downloads\tp-kubernetes> kubectl get pods --show-labels
NAME                     READY   STATUS    RESTARTS   AGE     LABELS
whoami-c6f799d48-j74bd   1/1     Running   0          2m25s   app=whoami,pod-template-hash=c6f799d48
whoami-c6f799d48-l28pw   1/1     Running   0          2m25s   app=whoami,pod-template-hash=c6f799d48
whoami-c6f799d48-pdc6j   1/1     Running   0          2m25s   app=whoami,pod-template-hash=c6f799d48
```



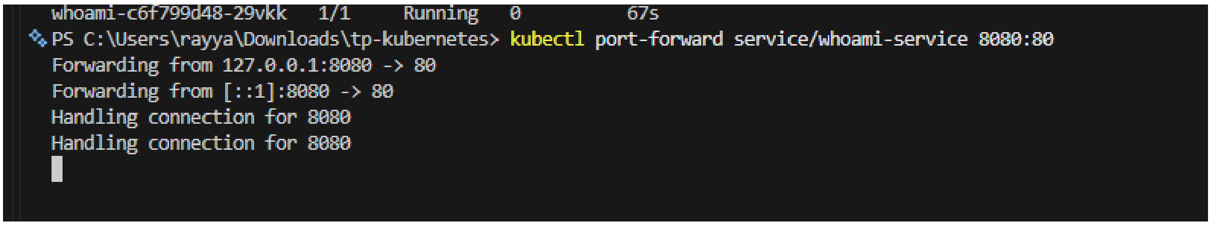
```
PS C:\Users\rayya\Downloads\tp-kubernetes> kubectl delete pod whoami-c6f799d48-j74bd
pod "whoami-c6f799d48-j74bd" deleted from demo namespace
```



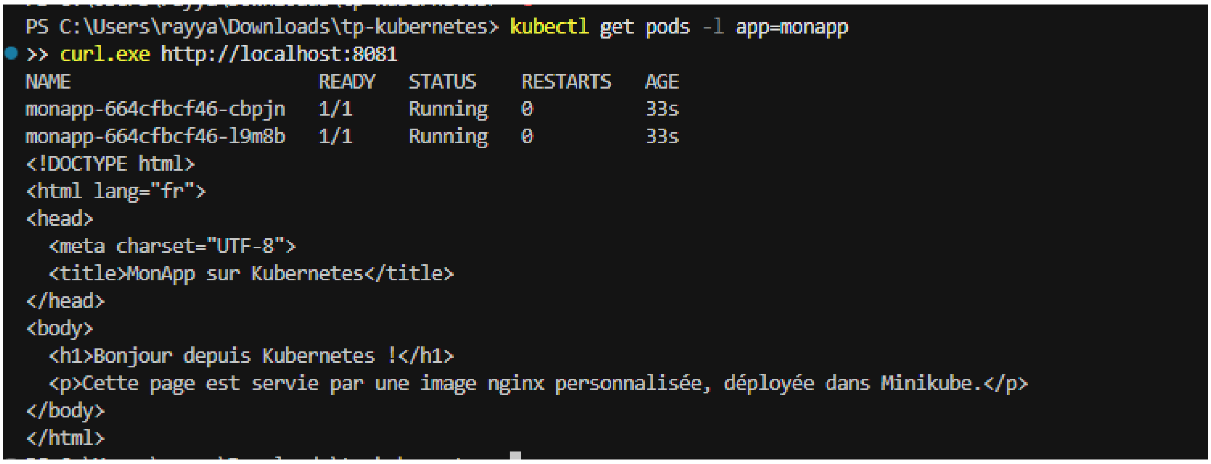
```
PS C:\Users\rayya\Downloads\tp-kubernetes> kubectl delete pod whoami-c6f799d48-j74bd
pod "whoami-c6f799d48-j74bd" deleted from demo namespace
PS C:\Users\rayya\Downloads\tp-kubernetes> kubectl get pods
NAME                     READY   STATUS    RESTARTS   AGE
whoami-c6f799d48-29vkk   1/1     Running   0          37s
whoami-c6f799d48-l28pw   1/1     Running   0          4m23s
whoami-c6f799d48-pdc6j   1/1     Running   0          4m23s
PS C:\Users\rayya\Downloads\tp-kubernetes> 
```



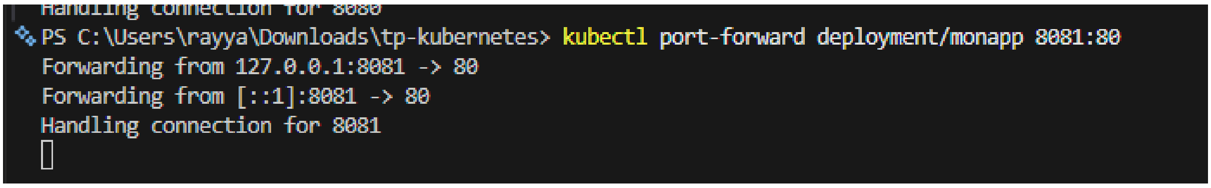
```
PS C:\Users\rayya\Downloads\tp-kubernetes> curl.exe http://localhost:8080
Hostname: whoami-c6f799d48-48vbb
IP: 127.0.0.1
IP: ::1
IP: 10.244.0.10
IP: fe80::78c7:ccff:fe47:ae46
RemoteAddr: 127.0.0.1:55674
GET / HTTP/1.1
Host: localhost:8080
User-Agent: curl/8.21.0
Accept: */*

PS C:\Users\rayya\Downloads\tp-kubernetes> curl.exe http://localhost:8080
Hostname: whoami-c6f799d48-48vbb
IP: 127.0.0.1
IP: ::1
IP: 10.244.0.10
IP: fe80::78c7:ccff:fe47:ae46
RemoteAddr: 127.0.0.1:46864
GET / HTTP/1.1
Host: localhost:8080
User-Agent: curl/8.21.0
Accept: */*
```



```
whoami-c6f799d48-29vkk   1/1     Running   0          67s
PS C:\Users\rayya\Downloads\tp-kubernetes> kubectl port-forward service/whoami-service 8080:80
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80
Handling connection for 8080
Handling connection for 8080
```



```
PS C:\Users\rayya\Downloads\tp-kubernetes> kubectl get pods -l app=monapp
>> curl.exe http://localhost:8081
NAME                      READY   STATUS    RESTARTS   AGE
monapp-664cfbcf46-cbpjn   1/1     Running   0          33s
monapp-664cfbcf46-19m8b   1/1     Running   0          33s
<!DOCTYPE html>
<html lang="fr">
<head>
  <meta charset="UTF-8">
  <title>MonApp sur Kubernetes</title>
</head>
<body>
  <h1>Bonjour depuis Kubernetes !</h1>
  <p>Cette page est servie par une image nginx personnalisée, déployée dans Minikube.</p>
</body>
</html>
```



```
Handling connection for 8080
PS C:\Users\rayya\Downloads\tp-kubernetes> kubectl port-forward deployment/monapp 8081:80
Forwarding from 127.0.0.1:8081 -> 80
Forwarding from [::1]:8081 -> 80
Handling connection for 8081
```